



Tackling a Hidden Universe of Uncertainty through Veridical Data Science

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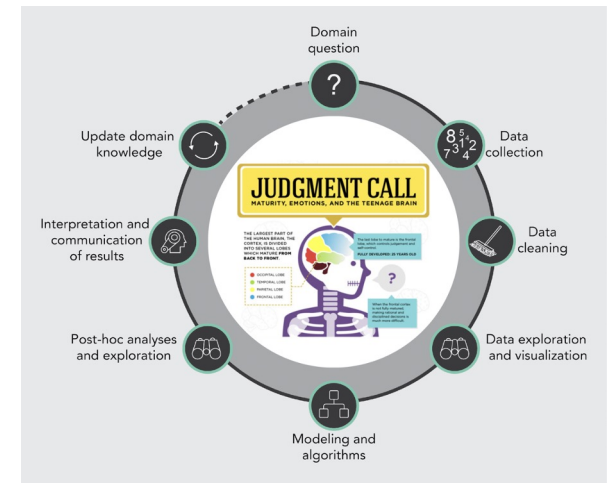
Le Cam Lecture, JDS

June 2, 2026

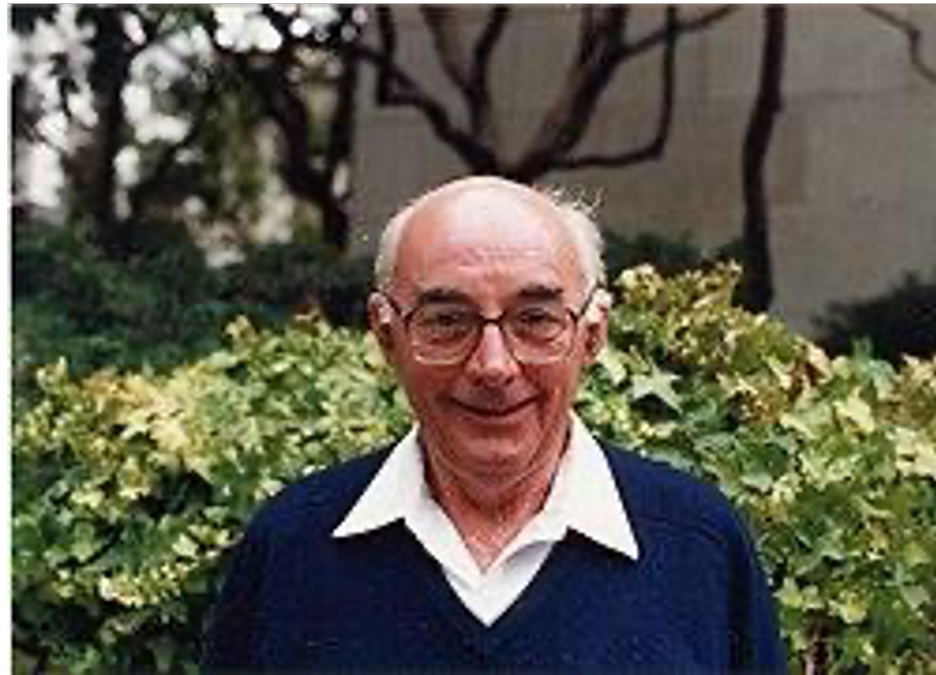
What does “veridical” mean in VDS?

Veridical means “truthful” in two ways in VDS:

1. It seeks truth in data conclusions
1. It is truthful to the data science life cycle (DSLCL)



Le Cam in 1987 when I was his student



1924 - 2000

All the quotes are from Yang (2002, AoS) "**LUCIEN LE CAM 1924–2000**"

Le Cam before Berkeley

Clermont-Ferrand (1942-1944)

Lycée

Eléments de Mathématique by Bourbaki

University of Paris (1944-1945)



Electricité de France (1945-1950) where he met Neyman

Le Cam's Theoretical Contributions in 7 years after his PhD in 1953 until Full Prof.

- Extension of Wald's decision theory
- Weak convergence of stochastic processes
- Locally asymptotic normal (LAN) theory and contiguity
- Deficiency and distance between two experiments
- Le Cam's distance

Probability Theory Contributions

- First usable definition of (bounded) Radon measures on completely regular spaces (tightness, τ -smooth and σ -smooth in use in mathematics)
- Approximations for the Poisson binomial distribution
- Several new inequalities in problems of approximation of distributions of sums by infinitely divisible ones

In Le Cam's words

Le Cam acknowledged in his book (1986):

“The ideas and techniques used reflect first and foremost the influence of **Abraham Wald**'s writings. Another influence was that of **Jerzy Neyman**, who asked a variety of questions but who also promoted my academic career. Some other easily discernible influences are those of **Jaroslav Hájek** and **Charles Stein**. Not so visible, but indispensable, were the teachings of **Etienne Halphen**, who attempted (without success) to convert us to the **Bayesian creed** long before it became fashionable, but who also **taught us a great deal about the interplay between theory and practice.**”

Interdisciplinary work in cancer research

- Neyman's interest got Le Cam into cancer research.
- "He was invited to participate in research in a clinical trial of a group of young children with osteosarcoma, including his daughter ... a series of papers in medical journals. This highly successful clinical trial was reported in the press."
- He also worked on "effects of radiation on living cells (with unpublished work) and sodium channel modeling."
- He believed that "it is hard to do applied work in which one has to worry about the nitty-gritty."

Meeting Le Cam

In 1984, I was in the China-US Chern Math Exchange Program; interviewed by professors sent by AMS including Fields Medalist T. S. Yau.

Switched from pure math to statistics PhD application with a rec. letter from AMS. .

Le Cam was the admissions chair at UCB and admitted me with a Berkeley Fellowship.

Summer of 1985, Le Cam visited China with Grace Yang.

I arrived in Berkeley in Aug. 1985.

Abstract Le Cam

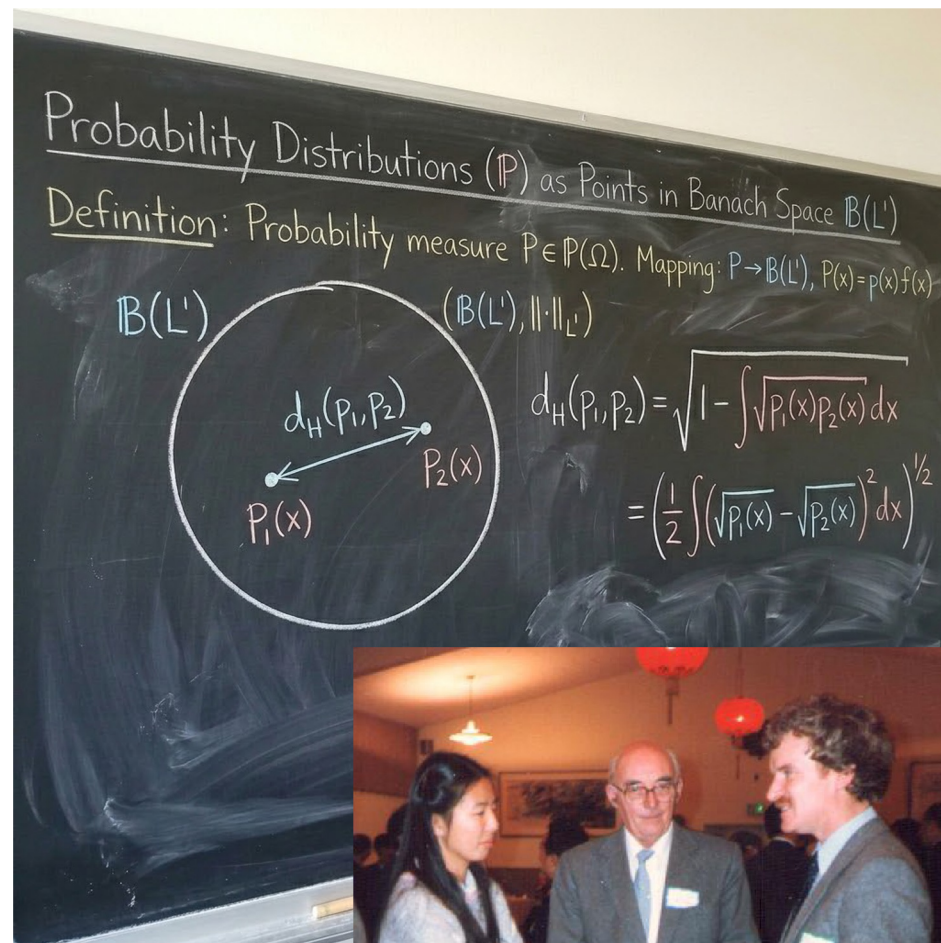
In his Stat 217 class on asymptotic theory TWICE:

A probability distribution is a point in a Banach space!

Hellinger distance makes it Hilbert!

Takeaway:

Abstraction can illuminate.



Thanks to Gemini.

Practical Le Cam

An imperfect recollection of a conversation in the late 80's:

Bin: what do you do with your asymptotic theory when you do cancer research?

Le Cam: I throw them out of the window.

Takeaway:

Be clear-minded about when something is useful and when it is not.

My work related directly to Le Cam

I persuaded Le Cam to take me on as his student on the condition that I found a topic myself, which I did in his big book – extending VC-class theory to dependent processes. But Le Cam thought Massart's Thesis in 1988 did what I did. So I decided to find something else to work on and Terry Speed came.

David Pollard later pointed out that I did the VC-class version and Massart did the bracket-entropy condition version.

So I put it in my thesis and published it in AoP with much help with Le Cam on dealing with measurability issues – but he refused to be a co-author on the paper.

The Annals of Probability
1994, Vol. 22, No. 1, 94–116

RATES OF CONVERGENCE FOR EMPIRICAL PROCESSES OF STATIONARY MIXING SEQUENCES¹

BY BIN YU

Assouad, Fano, and Le Cam

Bin Yu¹

Festschrift for Lucien Le Cam

Le Cam: a great human being with 38 PhD students

“A teacher and mentor: a scholar without ambition”

“made no distinction between his and other professors’ students.””His generosity and kindness nurtured numerous scholars around the world.”

“students’ protector”

“Taking a stand”

“Lucien was very modest about his own achievements but took a strong stand for what he believed in.”

“Rules are supposed to help, not to hinder.” – Le Cam

Thanksgiving dinner in my house for my students, postdocs, and new colleagues, inspired by Lucien



For my first thanksgiving in the US, Lucien invited me to his house...

2026 is the age of AI

We are in an AI Reproducibility Crisis in Science

Science

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Artificial intelligence faces reproducibility crisis

Unpublished code and sensitivity to training conditions make many claims hard to verify.

[MATTHEW HUTSON](#) [Authors Info & Affiliations](#)

SCIENCE • 16 Feb 2018 • Vol 359, Issue 6377 • pp. 725-726 • [DOI: 10.1126/science.359.6377.725](https://doi.org/10.1126/science.359.6377.725)

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NEWS FEATURE | 05 December 2023

Is AI leading to a reproducibility crisis in science?

Scientists worry that ill-informed use of artificial intelligence is driving a deluge of unreliable or useless research.

By [Philip Ball](#)

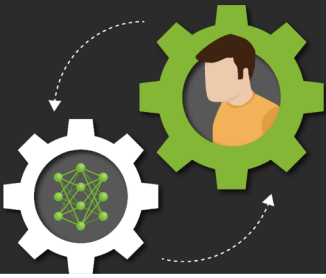
LLMs are doing data analysis, like it or not

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Technical Blog

Agentic AI / Generative AI English

Build an LLM-Powered Data Agent for Data Analysis



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TDS Archive

How LLMs Will Democratize Exploratory Data Analysis

Or, When you feel your life's too hard, just go have a talk with Claude



Ken Kehoe

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15 min read · Jun 9, 2024

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How does statistics thrive in the Age of AI?

“According to Darwin’s *Origin of Species*, it is not the most intellectual of the species that survives; it is not the strongest that survives; but the species that **survives** is the one that is able best **to adapt and adjust to the changing environment** in which it finds itself.”

– Megginson, L. C. (1963)

Statistics thrives by adapting to the changing environment...

“Statistics is the science of learning from data, and of measuring, controlling and communicating uncertainty.”

– ASA

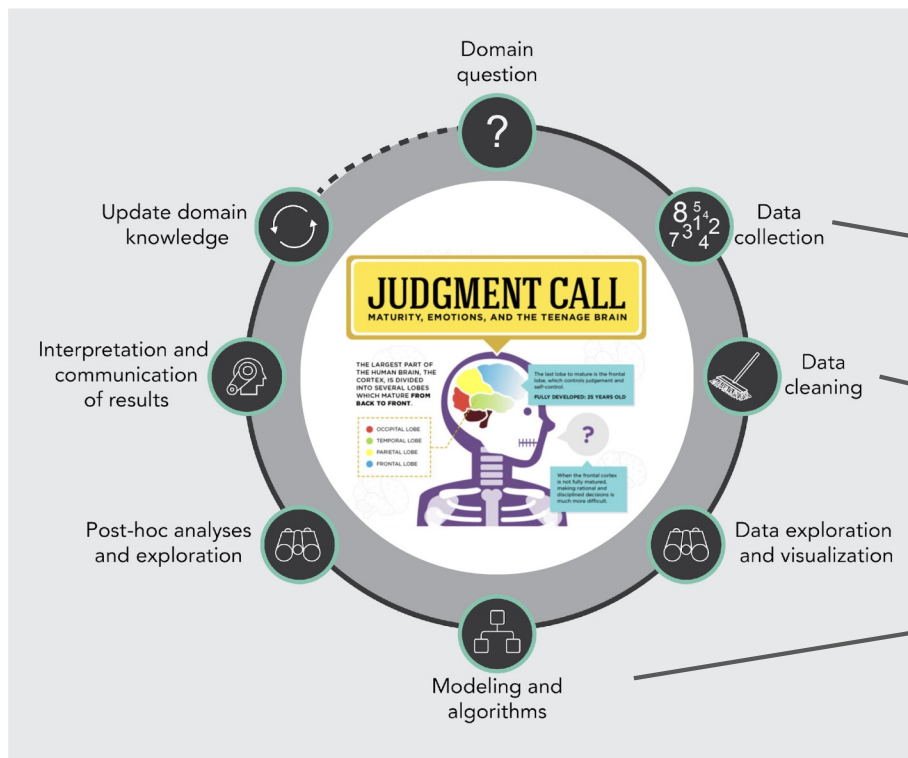
Statistics is at an **inflection point**...

Goal of Statistics in the Age of AI:

To provide **data evidence in context** for trustworthy conclusions and decisions, which rely on an entire DSLC.

Trustworthy conclusions rely on a full account of sources of uncertainty.

Data Science Life Cycle (DSLCL)



Human judgment calls in every step and they create missing uncertainty:

What choices were made while collecting data?

How was the data cleaned?

Modeling choices

A DSLCL creates uncertainty in every step!

Box (1979). Cox and Snell (1981), Nelder (1991)....

Image credits: R. Barter and toronto4kids.com

A hidden uncertainty source: problem formulation

Chan-Zuckerberg Biohub Intercampus Award (UCB, Stanford, UCSF):
finding genetic drivers of a heart disease HCM using UKBB data.

Human judgment calls made a difference:

1. Using HCM labels: can't beat random labels.
1. New responses/labels from **MRI images** yielded gene or gene-gene interaction prioritization list with **80% validated by wet-lab experiments** (4 out of 5).

See Wang, Tang, ..., Y. and Ashley (2025)
in *Nature Cardiovascular Research*, with commentary.

Another hidden uncertainty source: data cleaning

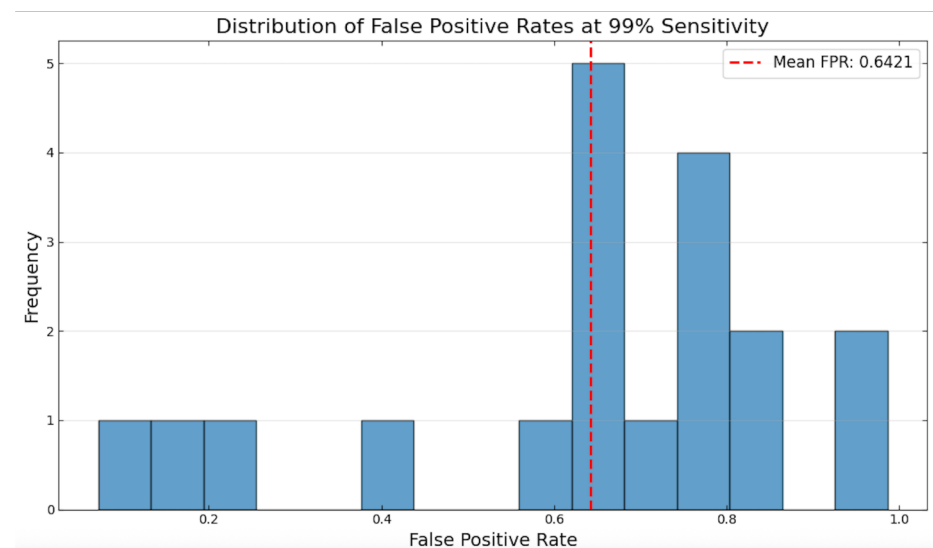
MA class project (stat 214), Spring 2025, UC Berkeley

- Goal: clinical decision rule on CT-scan or not for pediatric patients (with traumatic brain injuries), using clinical variables and labels
- **18 indep. students:** the same raw data, same data cleaning guidelines

At 99% sensitivity, estimated **false positive rates** range from **7.3%** to **98.7%**

– a **90% difference!**

Recommendation: keep at least **two cleaned datasets** and run down the pipeline. If final results differ a lot, argue for one or interpret the stable part.



Another uncertainty source: analyst/model choices

Social science paper in PNAS (2022) (a similar paper from biologists)

Observing many researchers using the same data and hypothesis reveals a hidden universe of uncertainty

[Nate Breznau](#)  , [Eike Mark Rinke](#) , [Alexander Wuttke](#) ,  +162, and [Tomasz Żółtak](#)  [Authors Info & Affiliations](#)

Edited by Douglas Massey, Princeton University, Princeton, NJ; received March 6, 2022; accepted August 22, 2022

“... **Seventy-three independent research teams** used identical cross-country survey data to test a prominent social science hypothesis... **teams’ results varied greatly, ranging from large negative to large positive effects** of immigration on social policy support.”

Patterns (2023)

Leakage and the reproducibility crisis in machine-learning-based science

Highlights

- Data leakage is a flaw in machine learning that leads to overoptimistic results

Authors

Sayash Kapoor, Arvind Narayanan

Errors and their propagation erode science: errors from initial 41 papers to 648 papers.

Veridical Data Science (VDS)
formally addresses hidden uncertainties

Veridical data science (VDS) draws on my research/teaching

I live in both the world of
illuminating abstraction/theory



<https://lifeology.io/blog/2020/07/27/the-spherical-cow-an-insight-into-science/>

and the world of practice



I now work as a bridge between the two worlds.

<https://photos.com/featured/cow-in-field-low-angle-robax.html?srsId=AfmBOoolxDXoQM3yKMMjN9nJ4atS0rgVGQIm IKCCZGHK55NXsvzQOsC>

Research over time: three periods

1. **BS in Pure Math, Peking University (thesis related to PDE)**
PhD in Statistics, UC Berkeley (thesis: theoretical machine learning and MDL theory) Rate-distortion theory; MCMC convergence analysis and diagnosis; Signal processing (wavelets); Network tomography, audio compression, **two review papers on MDL (Bell Labs)**



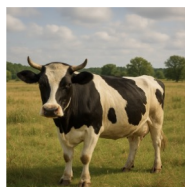
1. **Machine learning theory (bagging, boosting, Lasso, spectral clustering)**



Remote sensing (cloud detection, aerosol retrieval)
Neuroscience
(movie reconstruction, v4 characterization using DL)
Genomics (fate map, enhancer prediction)
Precision medicine (ER CDR stress-testing)



1. Genomics, neuroscience, prediction medicine
Climate science (MJO prediction using DL)
Interpretable ML and DL (e.g. mechanistic interpretation)
Veridical data science (cost-effective hypothesis generation)
Trustworthy and safe AI (medical AI, cancer detection)



Deep learning Th
Inf-width, LoRA+

Transition from theory to practice?

A random walk with a drift...

Transition from theory to practice?

A random walk with a drift...

- **Curiosity** about domain sciences and **interest** in solving real-world problems through

good mentors, friends in bioinformatics, sitting in a genetics class
- Picking up **good judgment calls** on what is essential and what is not about a domain problem, from mentors passively by observing and actively by asking questions.
- Learn from students/postdocs in domain sciences by **embedding oneself in a lab** and helped them with statistics
- Today, I use **chatGPT, Claude, and Gemini...**
- **Theory training gives a habit of rigorous thinking**
- **Interpersonal skills are very important.**

Rest of the Talk

1. VDS with Predictability-Computability-Stability (PCS) + documentation

AI for science (PCS towards causality):

overview on case studies/new ML algorithms

1. PCS Uncertainty Quantification (UQ)
2. PCS-UQ Theory under a Simple Model
3. CLEAR: combining PCS-UQ with QR
4. Sanity checks for Agentic Data Science

1. PCS theoretical research
2. Current directions and resources

Veridical Data Science (VDS) for trustworthy DS and AI



Y. and Kumbier (2020)

PNAS

VDS is built on three core principles of data science for **every step** of the **data science life cycle** (DSLC):

(P)redictability [ML and Stats] (“**reality-check**”)

(C)omputability [ML, “R”]

(**S**)tability [expanding uncertainty sources with user defined perturbations]

Veridical Data Science

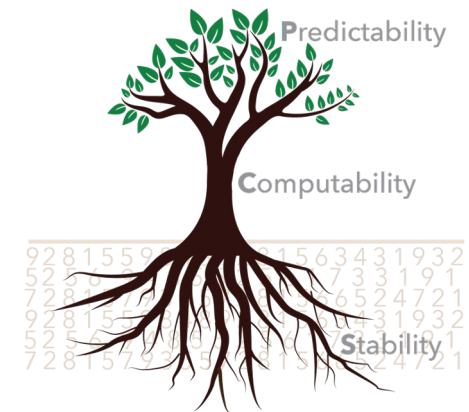


Image credit: R. Barter

PCS for VDS – a new paradigm

Unifying, synthesizing, and expanding on ML+Stats in **DSL**C.

Providing a unified language to **communicate/interpret** among **ourselves** and with **scientists or domain experts, managers**

Enabling trustworthy AI or AI safety

PCS documentation for **transparency** and **trust**

[on GitHub (^{JupyterNotebook} _{Quarto})]



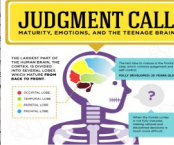
Reality



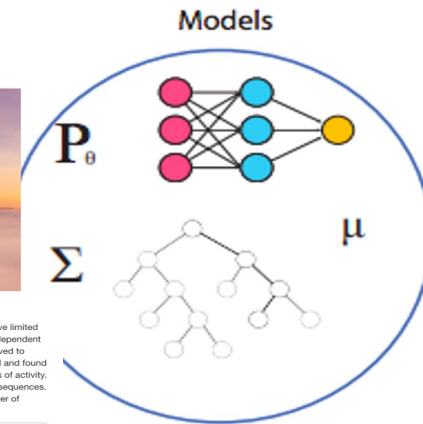
Stability formulation

Bootstrap sampling is a widely accepted perturbation understanding of the dependencies. However, sequer behavior that is possible to account for. In particular, confer robustness to regulatory processes (Hong, Hee that over 70% of loci they examined have anywhere fr To account for this potential dependency along the ge We define the stability of an interaction to be the prop bootstrap samples using the 3 proposed perturbation

```
# Block bootstrap for blocks of size 5 and
block5.tr <- makeblocks(gene.coords, idcs=
block5.tr <- makeblocks(gene.coords, idcs=
block5.tst <- makeblocks(gene.coords, idcs=test.id, size=5)
block10.tst <- makeblocks(gene.coords, idcs=test.id, size=10)
```



seful baseline for data where we have limited ce (i.e. nearby on the DNA) exhibit dependent m as "shadow enhancers" are believed to studied shadow enhancers in detail and found 316) with highly overlapping patterns of activity, urbations using blocks of 5 and 10 sequences. $j = 100$ RFs trained on an outer layer of



Mental Construct

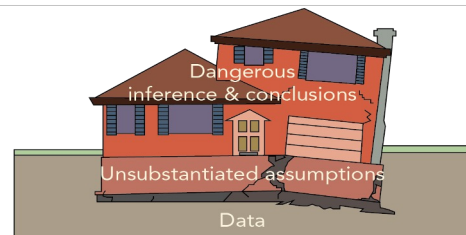


Image credits: Rebecca Barter

PCS documentation template: <https://yu-group.github.io/vdocs/PCSDoc-Template.html>

How to choose **perturbations** for “S”?

For **each step** of DSLC, there are **multiple reasonable choices determined by context**, possibly favored with different weights based on prior knowledge, and subject to resource constraints; there might also be **multiple stability metrics** for each perturbation.

Meta judgment calls are still needed; aggregations could improve final results.

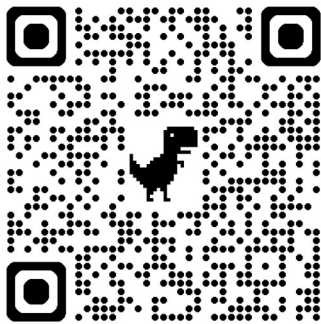
Record all human reasoning and judgment calls using **PCS documentation**.

Design issues in PCS implementation are research questions in PCS.

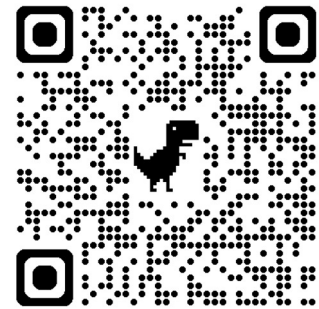
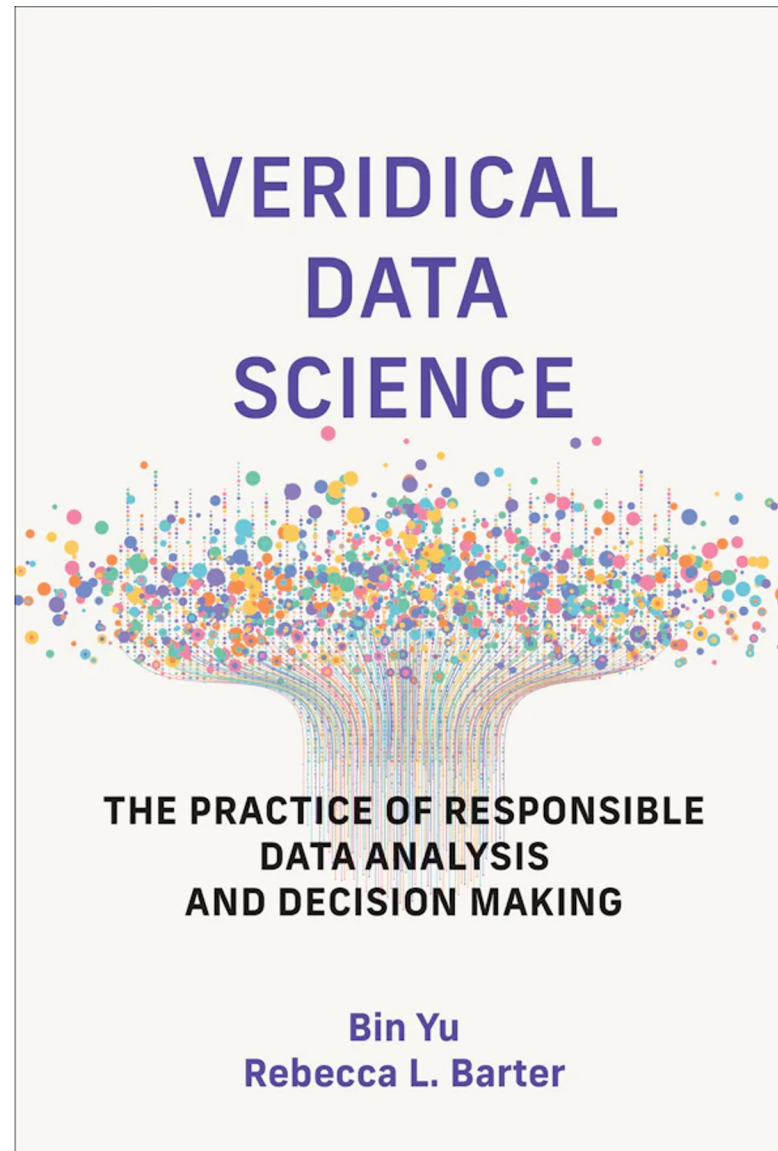
A related work is “Forking” by Gelman and Loken, 2014.



Bin Yu Rebecca Barter

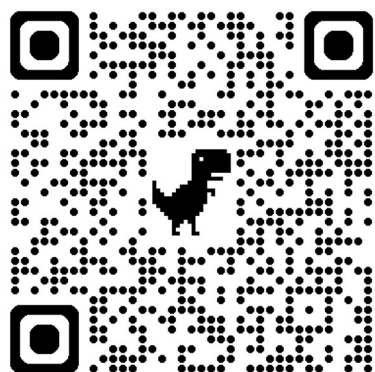


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vdsbook.com
Feb, 2024
Took 9 Years



MIT Press · '24
(ML Series)
Review in HDSR
by Benjaminis

Sept, 2025



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丽贝卡 L 巴特

真实数据科学

VERIDICAL DATA SCIENCE



负责任的数据分析与决策实践

THE PRACTICE OF RESPONSIBLE
DATA ANALYSIS AND DECISION MAKING

陈松蹊

中国科学院 院士

刘 军

美国国家科学院 院士

倾 / 情 / 推 / 荐

美国艺术与科学院、美国国家科学院双院院士郁彬教授：
三十年学术思想精粹！

[美] 郁彬 丽贝卡·L·巴特 (Rebecca L. Barter) 著
常象宇 贾金柱 刘汉中 吕晓玲 译 耿直 校

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[真实数据科学- 中国高校教材图书网](http://vdsbook.com)

AI for Science via PCS
reality check through “P”
varied uncertainty considerations via “S”

The PCS case studies to follow are about interpretable and reproducible scientific results guided by **PCS – prerequisite for interpretability.**

Interpretability is critical for trustworthy AI.

Causality Spectrum and PCS

Mechanistic
Individual level

...

Average effect
Group level

Stable, replicable

Effect depends on the group
Stability implicit in causal
inference: e.g. SUTVA

PCS works towards causality:

Predictability + stability (+ computability)



interpretable and cost-effective hypothesis prioritization

AI for science: PCS hypotheses + experiments  **causal conclusions**

AI for Science Case Studies with New ML Methods (finding appropriate “S” to go “P” in context)

AI for Science Case Studies with New ML Methods (finding appropriate “S” to go “P” in context)

Iterative random forests (iRF) developed in context of **developmental biology** (past experiments validate 80% of 20 **causal epistatic genetic interactions**) (Basu et al, 2018, *PNAS*)

epiTree hypothesis testing for epistatic interaction detection based on iRF (Behr et al, 2024, *PLOS One*)

lo-siRF developed in context of **precision medicine** (**heart disease HCM**) to find **two causal epistatic genetic interactions** (ran 5 experiments, 4 found to be causal) (Wang et al (2025), *Nature Cardiovascular Res.*, with commentary) [SANO Genetics Podcast Interview](#)

Generative Causal Test (GCT) in context of **neuroscience** or understanding language brain regions (ran fMRI experiments) (Antonello et al, 2025, revised for *Nature Neuroscience*)

AI for Science Case Studies with New ML Methods (finding appropriate “S” to go “P” in context)

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New ML method **PCS Ranking** led to **cost-effective prostate cancer detection** (Tang et al, 2025, *Cancer Biomarkers*)

New ML method **NESS** to **improve t-SNE and UMAP** for smooth structures in **single cell** data (Ma et al, 2025, in revision for *Patterns*). 44

PCS Uncertainty Quantification

PCS UQ for regression and classification



R. Barter*

PCS regression perturbation interval (Ch. 13 of Yu-Barter book) and classification and comp. efficient PCS UQ for deep-learning (new)



Abhineet Agarwal*



Michael Xiao*



Boyu(Boris) Fan



Omer Ronen

* denotes equal contribution

“PCS-UQ: Uncertainty Quantification via the Predictability-Computability-Stability Framework”
<https://arxiv.org/abs/2505.08784> (revised for PNAS)

PCS Perturbation Interval

VDS book considers three sources of uncertainty in DSLC from

1. Data collection process (existing)

1. Data cleaning choices (new)

1. Pred-checked modeling choices (new)

PCS UQ relies on a finite collection of pseudo datasets or values of interest, as in bootstrap.

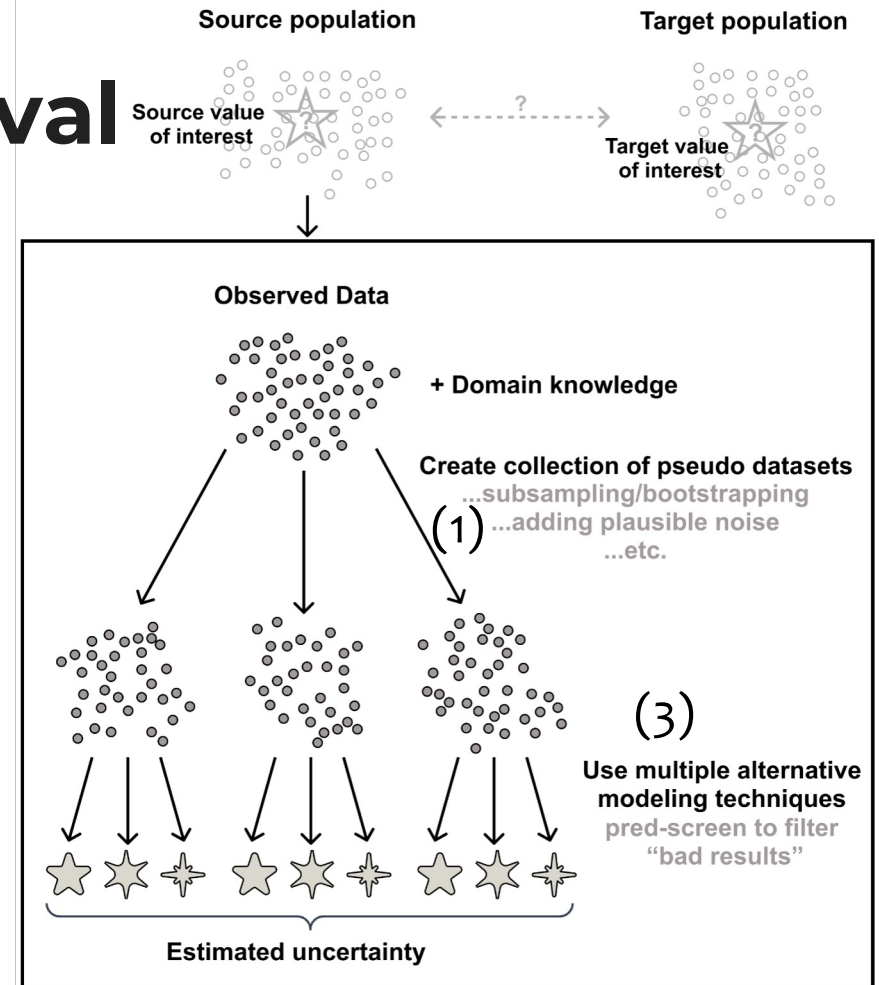
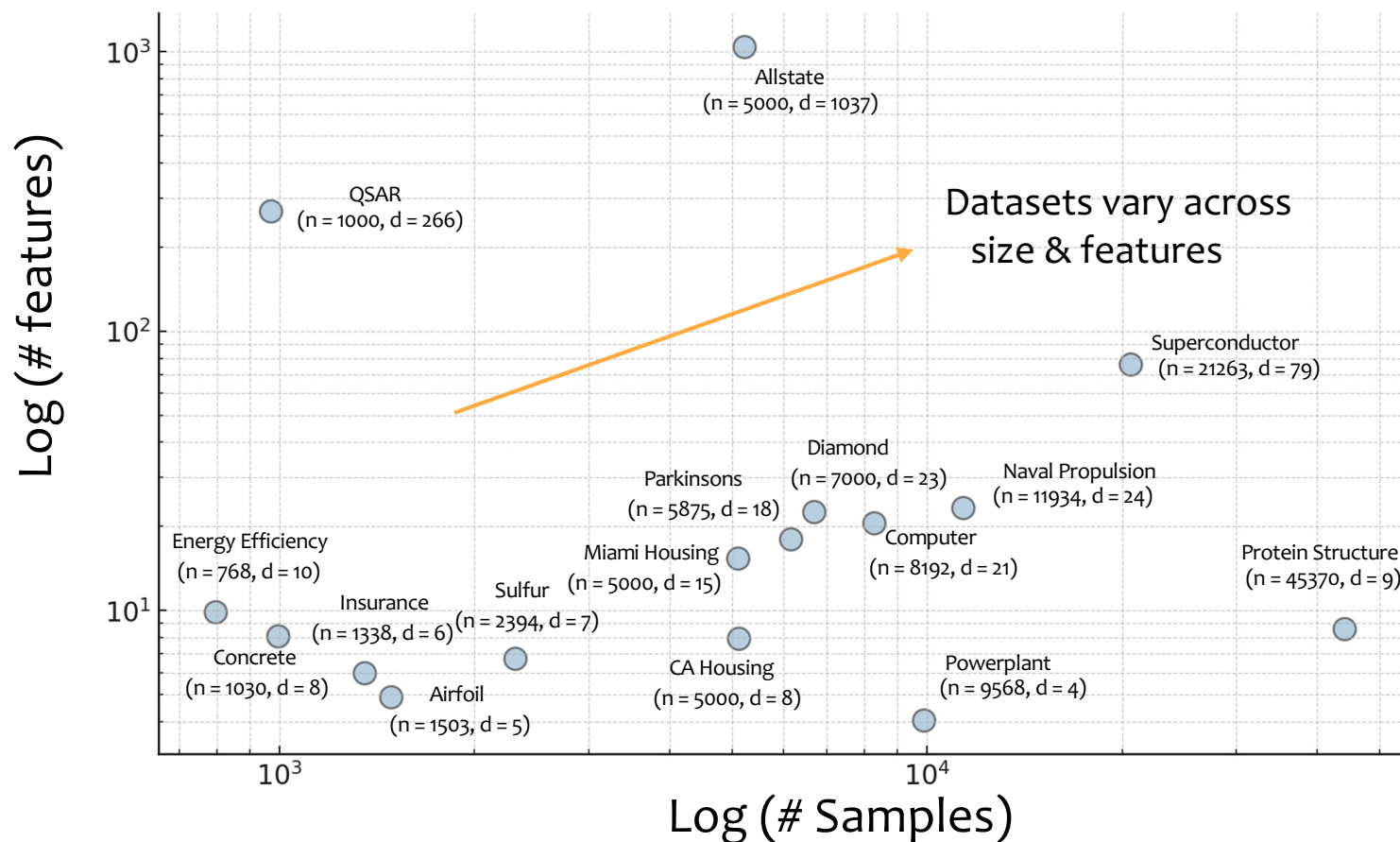


image credit: R. Barter

First(?) Benchmark for UQ: 17 Real-World Regression Datasets (no data cleaning uncertainty)



Comparison to conformal prediction (CP): split data into train and val. sets

PCS

- Data cleaning uncertainty allowed
- Uses **multiple** ML algorithms
- “**Pred-check**” to screen out bad alg.
- “Local” calibration on val. set via stability (**bootstrap** multiple pred-checked algorithms, data cleaning)
- **Multiplicative length calibration** to achieve empirical coverage on validation set under assumption that validation set is a good proxy to future data

Split Conformal

- No data cleaning uncertainty
- Uses **one** ML model
- No explicit model checking
- Global calibration using residuals
- **Constant length calibration**, to achieve coverage if exchangeability assumption holds between future data and current data

PCS Hyper-parameters

- Candidate models:
 - Linear: OLS, Lasso, Ridge, ElasticNet,
 - Bagging: Random Forests (RFs), ExtraTrees
 - Boosting: XGBoost, AdaBoost,
 - DL: Multi-layer Perceptrons (1 hidden layer)
- Top-1 model & 1000 bootstraps chosen via **synthetic simulations & 5 pilot datasets**

No contamination

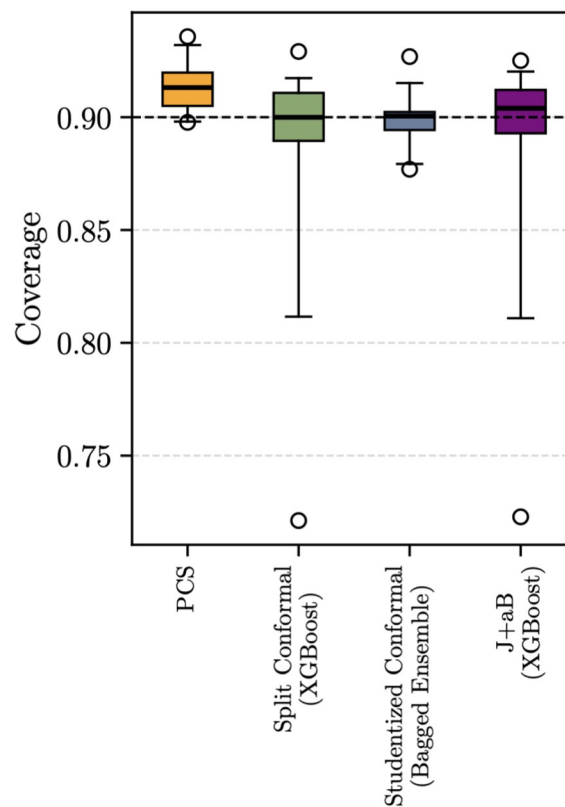
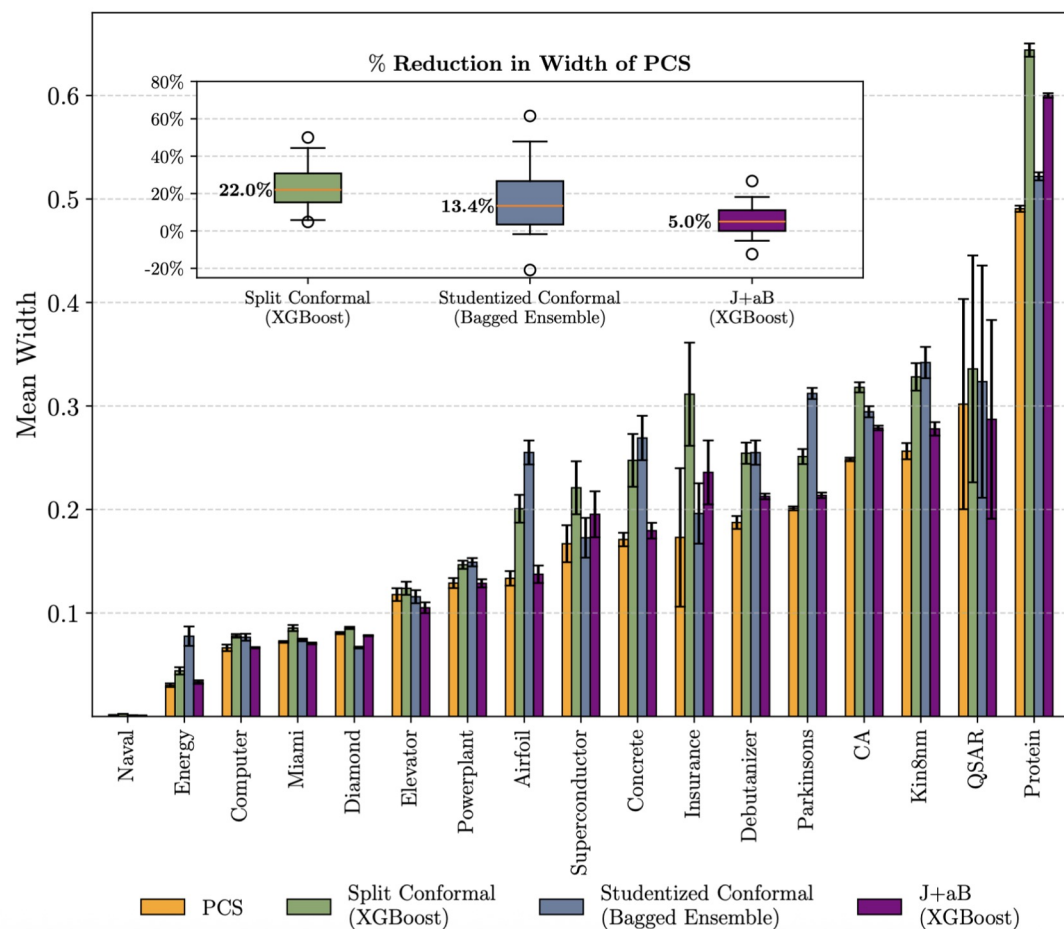
Conformal Hyper-parameters

- Candidate models: OLS, Lasso, Ridge, ElasticNet, Random Forests (RFs), XGBoost, ExtraTrees, Multi-layer Perceptrons (1 hidden layer)
- Try **all** candidate models and use **best** one for conformal over the 17 datasets in terms of test sets in our benchmark.

The CP methods are “oracle prediction-checked” over 17 datasets, while PCS-UQ prediction-checked on validation set for each of 17 datasets

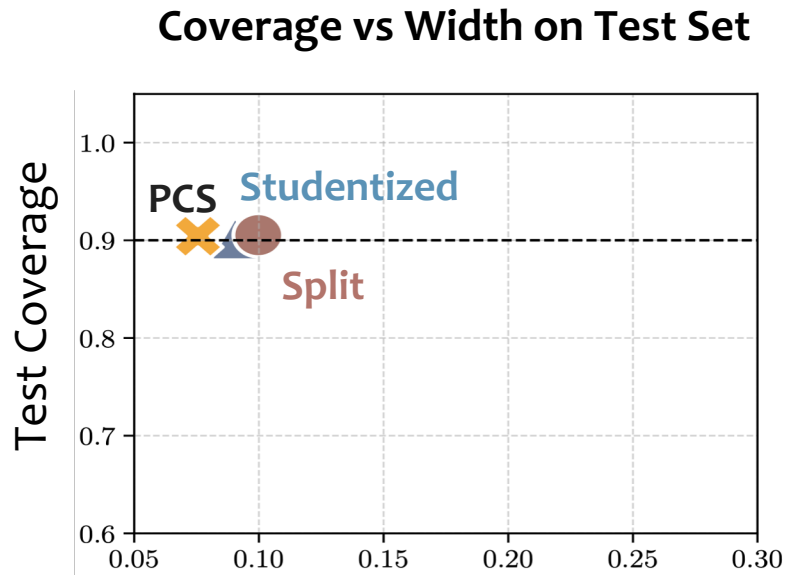
Width reduction by PCS-UQ (using OOB) (22%, 13%, 5%)

Marginal coverage all good, but split and J+aB vary a lot

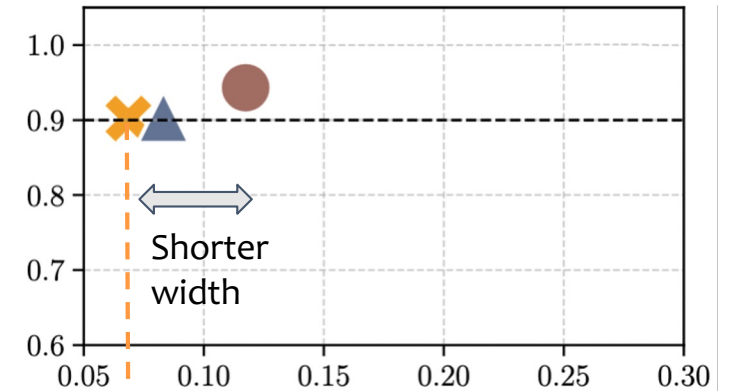


Subgroup results

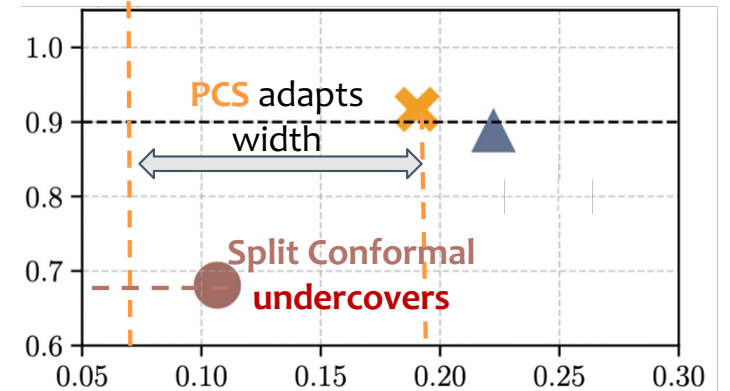
Miami Housing (n=13932, d=28)



Low
income



High
income



Takeaways:

1. PCS & Studentized adapt; Split + J+aB do not
2. PCS shorter width than Studentized

J+aB: best CP among 3 CPs compared

A Jackknife based CP, J+aB, uses bootstrap ensemble (Kim et al. 2020).

The “oracle pred-checked” J+aB gives similar marginal coverage and 5% worse width performance than PCS-UQ.

J+aB is **closely related to PCS-UQ since it uses bootstrap ensemble.**

However, **PCS-UQ outperforms J+aB on subgroup results.**

Beginning of convergence of CP and PCS ideas for UQ, although PCS is a broad framework for DSLC.

Recap: three components of PCS-UQ

- **Prediction-check** ensures predictive algorithms adequately capture reality
 - Better algorithms produce **tighter intervals**.
- **Bootstrap** captures inter-sample variability
 - Ensures **stability** and **marginal** coverage.
 - Simulates distribution of prediction of each individual sample -> **local uncertainty**.
- **Multiplicative calibration** adjusts interval widths adaptive to level of uncertainty
 - Gives consistently valid coverage across **subgroups**.

PCS-UQ: Multi-Class & DL Results

Takeaways across datasets

- **PCS** and **conformal** achieve **desired coverage** across 6 tabular datasets
- **PCS** reduces width over best conformal by **~20%**

Deep-learning (3 datasets)

- Provide approximation schemes to overcome computationally expensive bootstrap training
- **PCS fast approximation schemes outperform conformal variants**

Deep-Learning Experiments

		CIFAR 100			Caltech Birds			Imagenet-Small		
Method		Cov.	Av. Size	Time	Cov.	Av. Size	Time	Cov.	Av. Size	Time
APS		90.6	6.8	2	91.4	16.6	2	90.6	14.4	3
RAPS		90.8	6.5	2	91.1	12.8	2	90.5	11.2	3
TopK		92.3	8.5	2	91.6	17.6	2	92.3	13	3
PCS	Original	90.5	3.7	350	92.4	8.3	100	90.4	8.8	500
	Dropout	91.3	4.4	4	92.1	9.8	3	90.6	9.8	5
	Noise	91.1	4.2	3	91.9	9.4	3	90.5	9.6	5

Takeaways:

1. **Original PCS smallest size, and PCS approx. second best**
2. **PCS approx. are efficient**

PCS-UQ: Theory

PCS-UQ theory (on-going)

Our theoretical analysis focuses on **bootstrap** (which captures sampling variability) and **multiplicative calibration** (for adaptivity).

Co-authors



Anthony Ozerov



Yuetian Luo



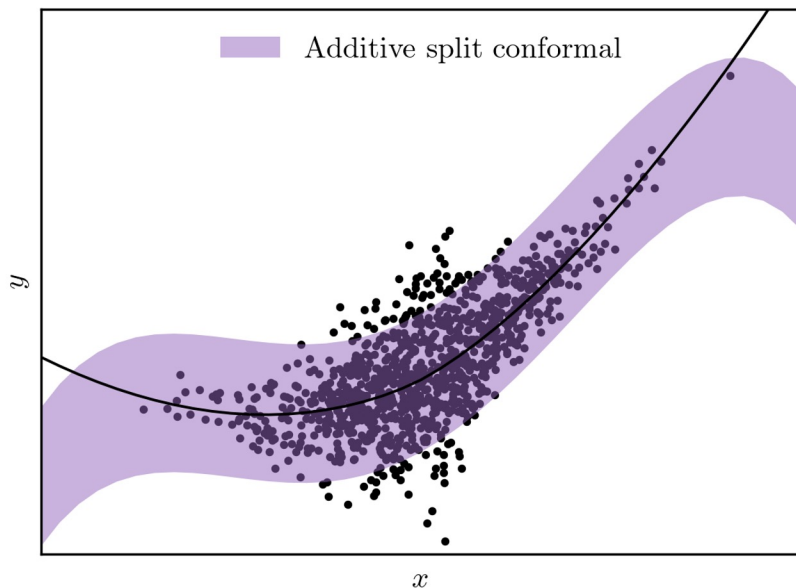
Minge Xie

The multiplicative calibration in PCS-UQ is a form of conformal prediction (Agarwal et al, 2026+)

Under exchangeability of a new test point with the calibration data, PCS-UQ achieves **marginal coverage**:

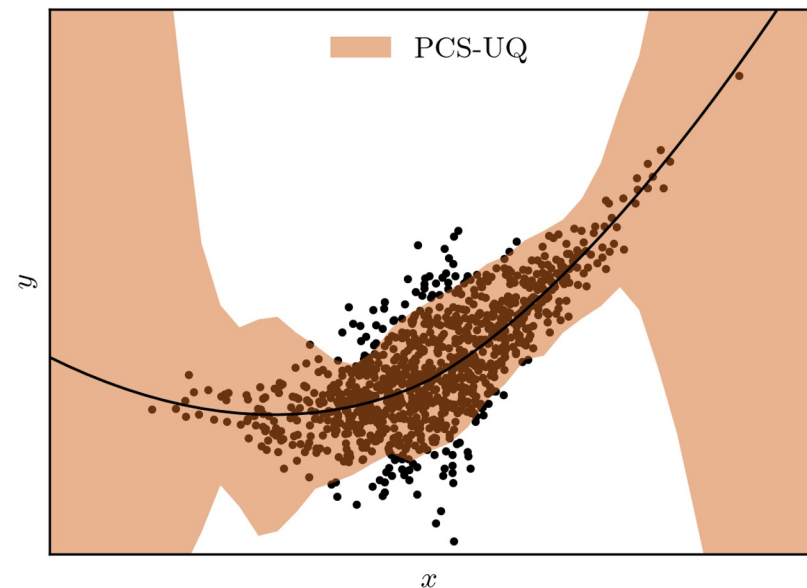
$$\underline{P(Y \in \hat{C}_{PCS}(X)) \geq 1 - \alpha}$$

“Additive” split conformal vs “Multiplicative” PCS-UQ



$$\hat{f}(x) \pm \gamma$$

← **Additive**
calibration
constant



$$\hat{f}(x) \pm \gamma q_\beta \left(\{ |\hat{f}_b(x) - \hat{f}(x)| \}_{b=1}^B \right)$$

Calibration constant that **multiplicatively scales**
bootstrap quantiles

Why does PCS-UQ get good conditional or subgroup coverage in 17 datasets?

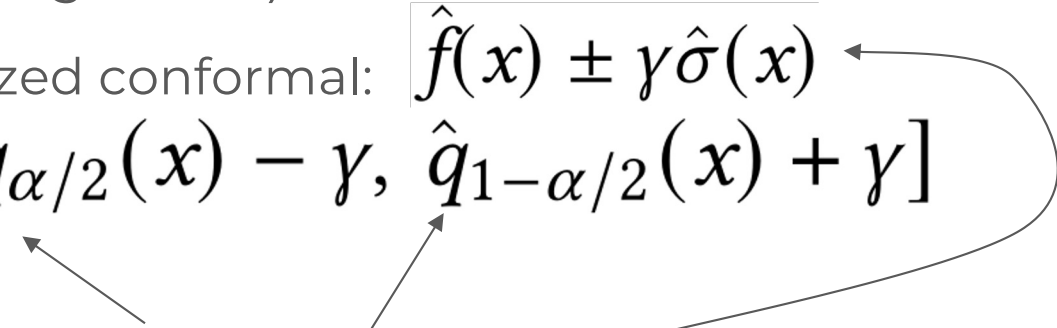
Theoretically PCS-UQ has no subgroup-conditional coverage guarantee.

In practice it works well, even better than methods that have a theoretical guarantee (studentized conformal; conformalized quantile regression)

Studentized conformal: $\hat{f}(x) \pm \gamma \hat{\sigma}(x)$

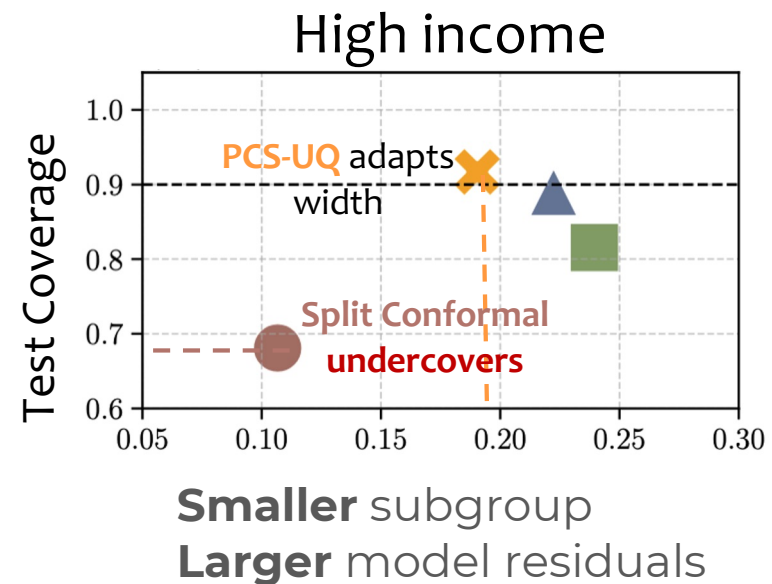
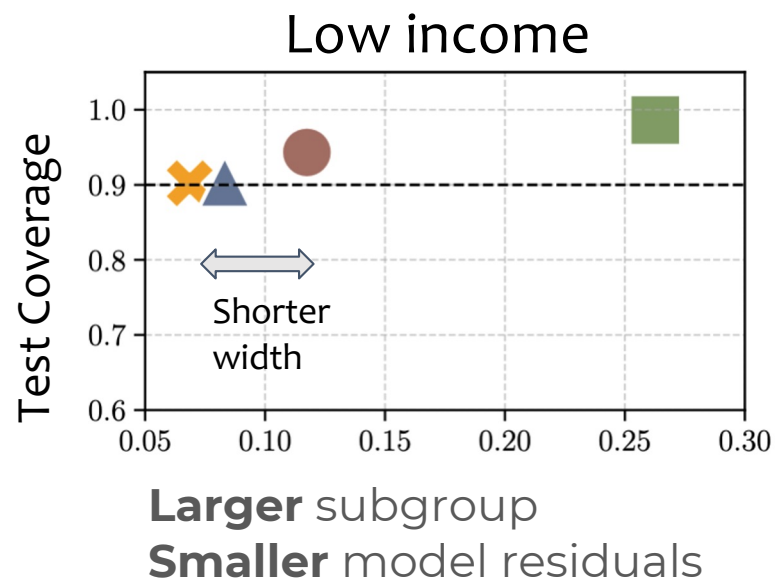
CQR: $[\hat{q}_{\alpha/2}(x) - \gamma, \hat{q}_{1-\alpha/2}(x) + \gamma]$

Hard to estimate!



The diagram consists of three arrows originating from the text 'Hard to estimate!'. One arrow points to the $\hat{\sigma}(x)$ term in the studentized conformal formula. Another arrow points to the $\hat{q}_{\alpha/2}(x)$ term in the CQR formula. A third, longer arrow points from the text to the $\hat{q}_{1-\alpha/2}(x)$ term in the CQR formula.

Two-group model: motivation



These examples suggest studying a **two-group model** where we vary the **size of the groups** and the **within-group variance**.

for training or fine-tuning using

Two-group model: asymptotic results

Consider two groups with different means and variances

$$\mathcal{X} = \{0, 1\}. X \sim \text{Ber}(p), Y|X=0 \sim N(\mu_0, \sigma_0^2). Y|X=1 \sim N(\mu_1, \sigma_1^2)$$

Asymptotic results (training and calibration data $\rightarrow \infty$, bootstraps $\rightarrow \infty$):

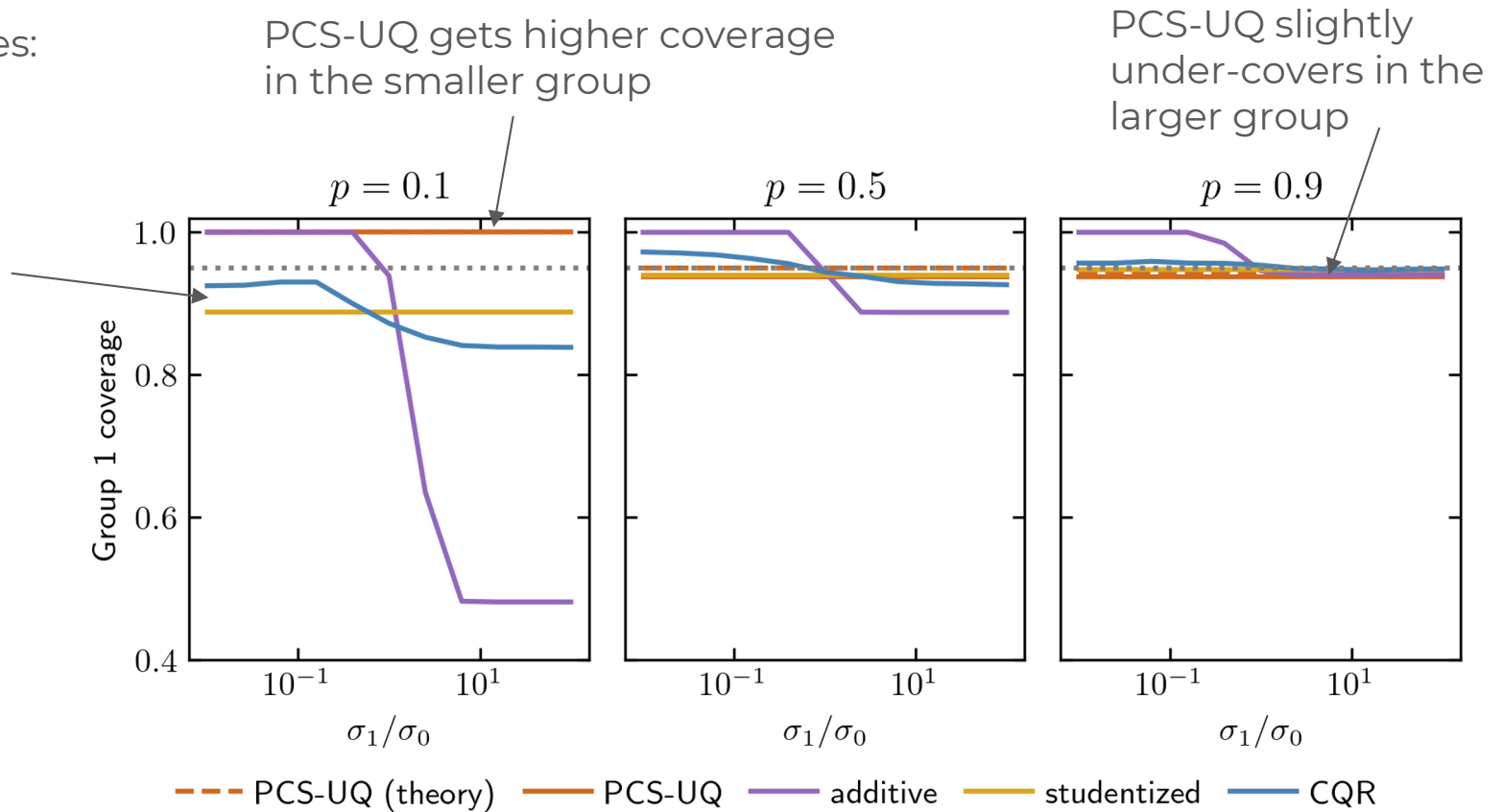
- PCS-UQ will attain **“close” to group-conditional coverage** in many settings of the DGP.
- Conditional coverage does not depend on σ_0, σ_1
- Coverage probability is higher in the smaller group.

Example: $p=0.9, \alpha=0.05 \rightarrow$ Group 0: 100%, Group 1: 94.4%

Abstraction/theory can illuminate.

Two-group model: finite-sample results

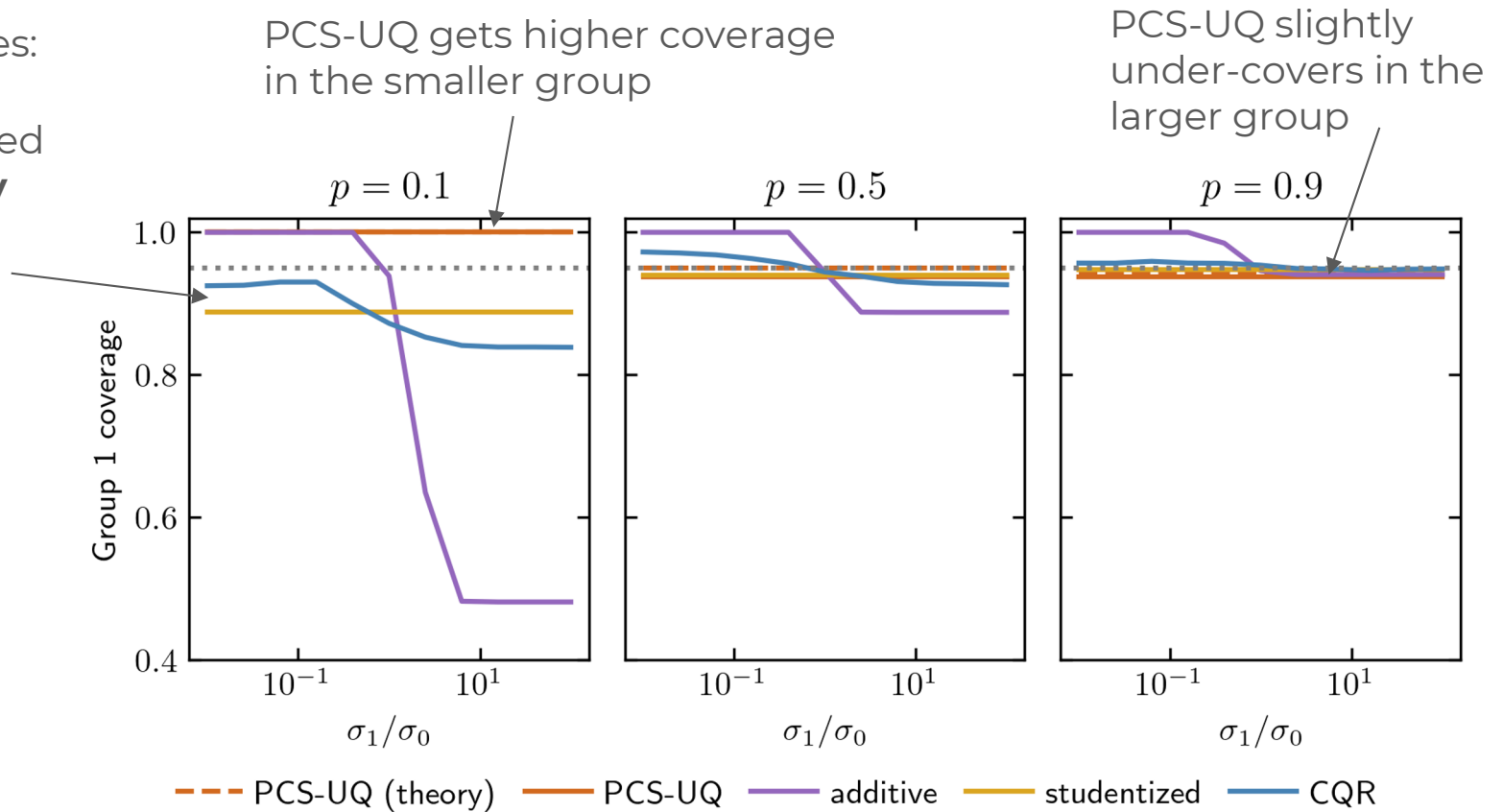
100 training samples:



Two-group model: finite-sample results

100 training samples:

CQR and studentized conformal **severely under-cover** in the smaller group.



Two-group model: finite-sample results

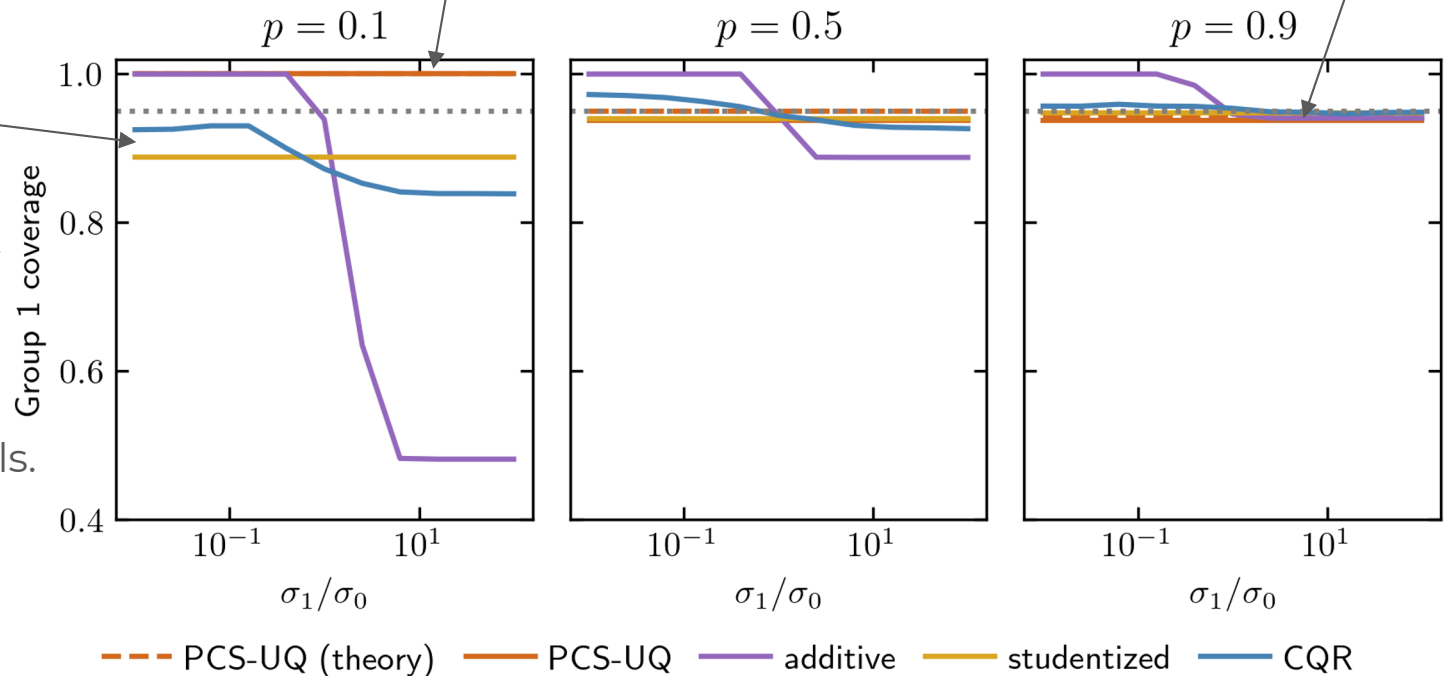
100 training samples:

CQR and studentized conformal **severely under-cover** in the smaller group.

Why? CQR requires estimating quantiles of $Y|X$, and studentized conformal requires estimating the s.d. of $Y|X$ or of model residuals. These are hard to estimate in finite samples!

PCS-UQ gets higher coverage in the smaller group

PCS-UQ slightly under-covers in the larger group ,



Takeaways from theoretical analysis of PCS-UQ

Putting aside the benefits of multiple models, screening, and sample efficiency, PCS-UQ obtains the following from its **multiplicative calibration** of a bootstrap ensemble:

1. PCS-UQ is a **new form of conformal prediction**.
2. Conformal methods estimate the quantiles or variance of the distribution $Y|X$ to obtain conditional coverage, but this is **difficult in finite samples**, even in a very simple model!
3. **PCS-UQ gets better conditional coverage** in finite samples than methods with a theoretical guarantee.

Le Cam used to say that the usefulness of asymptotics lies in its finite sample approximation properties. It doesn't approximate well here.

CLEAR: improving PCS-UQ with Quantile Reg

CLEAR: Calibrated Learning for Epistemic and Aleatoric Risk (ICLR, 2026)

CLEAR combines **PCS-UQ** and **Quantile Regression (QR)**;
scales them by **two calibration constants**;
reduces the width while maintaining the coverage.

Co-authors



Ilia Azizi*



Juraj Bodik*

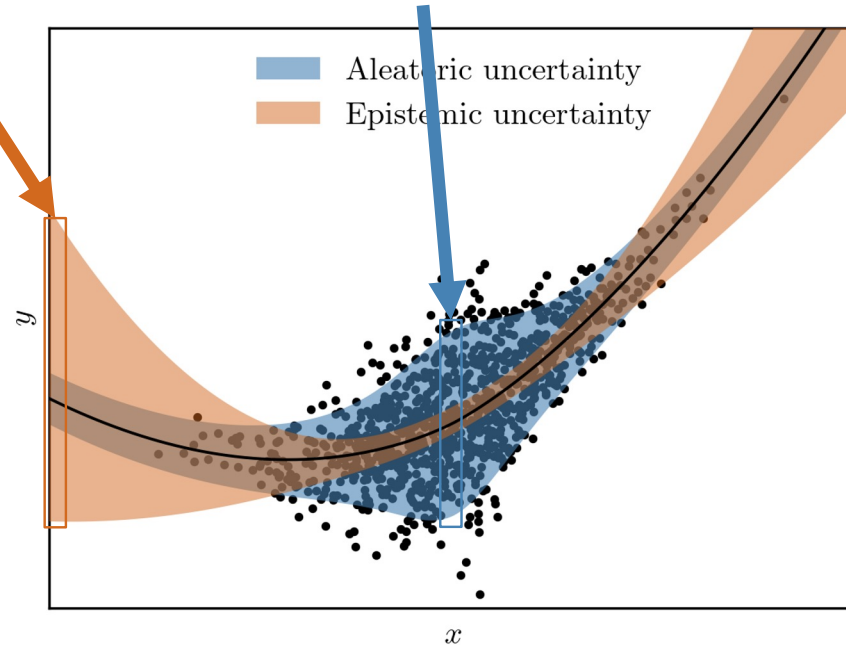


Jakob Heiss*

* denotes equal contribution

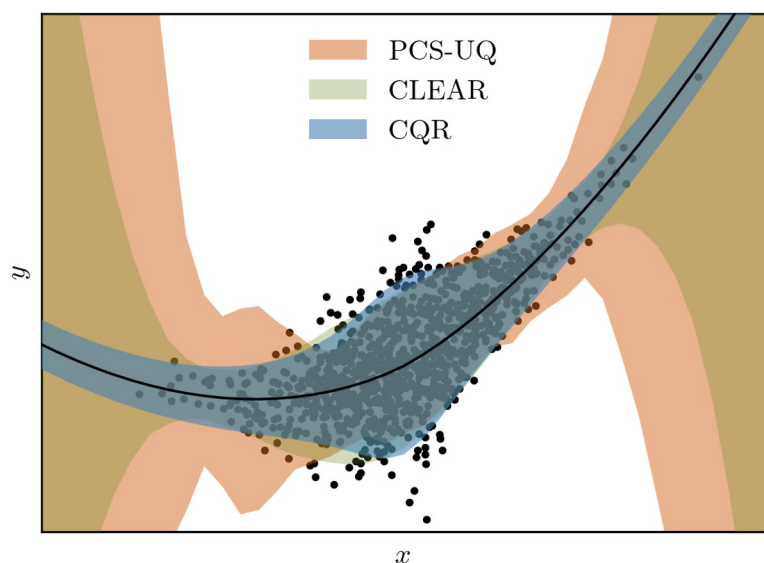
Terminology:

Epistemic vs Aleatoric



Reason	missing experience	missing context
Algorithms	Ensemble, PCS-UQ, GPR	Quantile Regression, ML for (μ, σ)

CLEAR: Calibrated Learning for Epistemic and Aleatoric Risk (Azizi et al., ICLR 2026)



PCS estimated via PCS-UQ

$$C = \left[\hat{f} - \underbrace{\gamma_1 \hat{q}_{\alpha/2}^{\text{ale}} - \gamma_2 \hat{q}_{\alpha/2}^{\text{epi}}}_{\text{estimated via Quantile Regression (QR)}}, \hat{f} + \underbrace{\gamma_1 \hat{q}_{1-\alpha/2}^{\text{ale}} + \gamma_2 \hat{q}_{1-\alpha/2}^{\text{epi}}}_{\text{estimated via Quantile Regression (QR)}} \right]$$

estimated via Quantile Regression (QR)

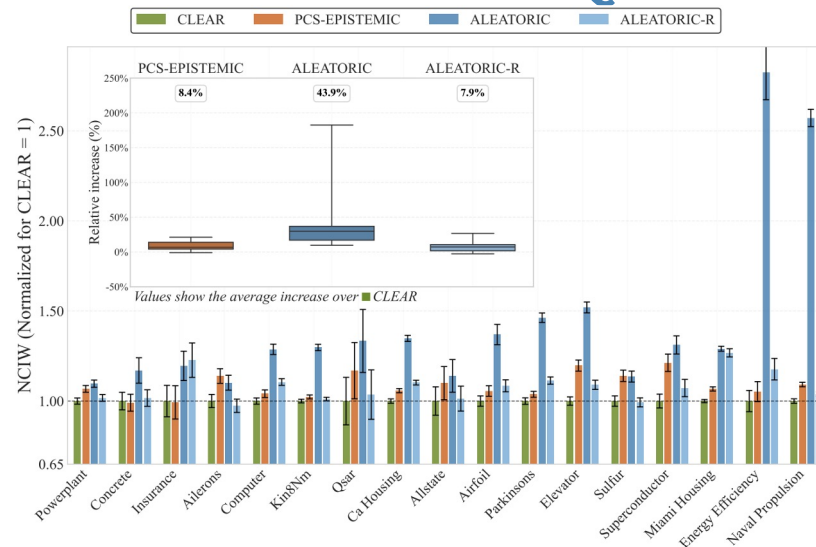
Calibrate 2 constants γ_1, γ_2 such that on the validation dataset:

- Marginal coverage is satisfied
- Quantile Loss is minimized

Evaluation on 17 datasets

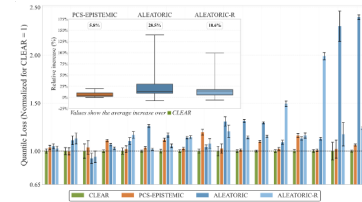
Maintaining the same target marginal coverage, **CLEAR** reduces the average interval **width**:

- **7%** compared to **PCS-UQ**
- **25%** compared to **Conformalized Quantile Regression (CQR)**



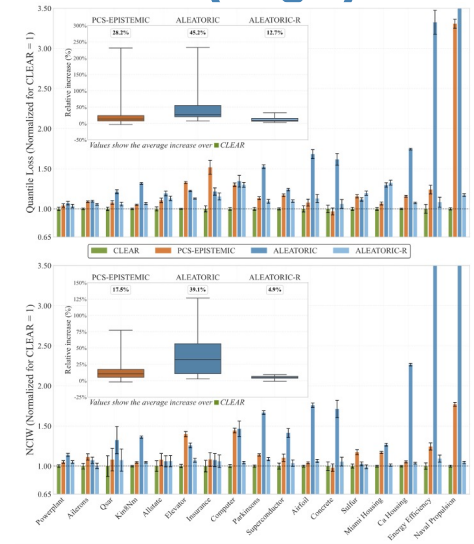
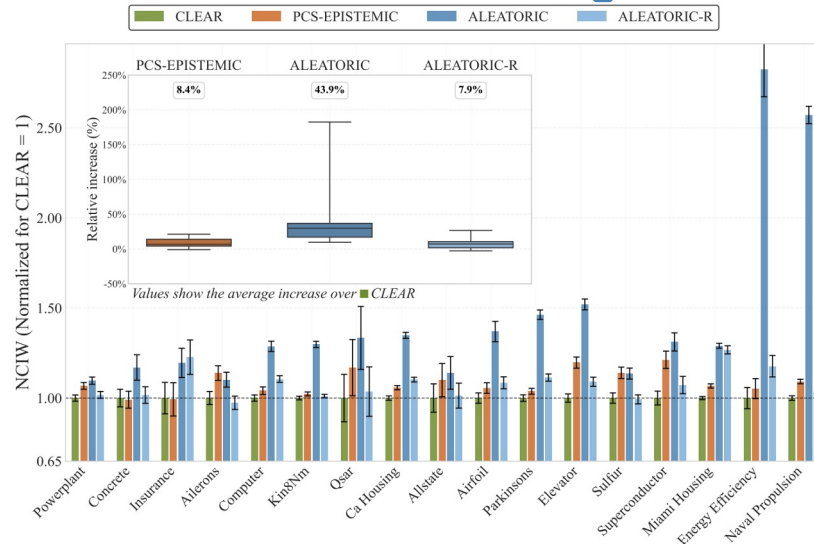
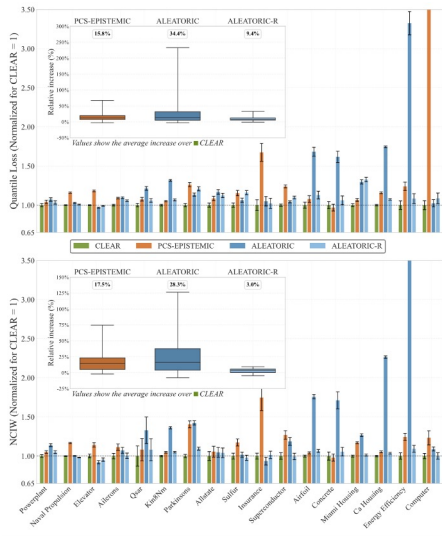
“S”: Similar results for different metrics and for different base models

Evaluation on 17 datasets



Maintaining the same target marginal coverage, **CLEAR** reduces the average interval **width**:

- **7%** compared to **PCS-UQ**
- **25%** compared to **Conformalized Quantile Regression (CQR)**



Extending the **same concept** (using two calibration constants) to **classification** yields **similar results**.

“**JUCAL**: Jointly **Calibrating Aleatoric** and **Epistemic** Uncertainty in Classification Tasks” (Arxiv 2026)

Co-authors:

Jakob Heiss*, Sören Lambrecht*, Josef Teichmann, Jakob Weissteiner, Hanna Wutte, Žan Žurič



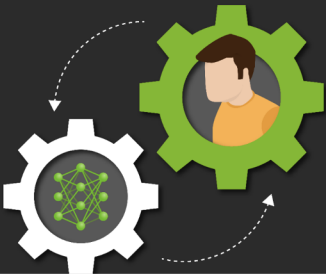
LLMs are doing data analysis, like it or not

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Technical Blog

Agentic AI / Generative AI English

Build an LLM-Powered Data Agent for Data Analysis



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How LLMs Will Democratize Exploratory Data Analysis

Or, When you feel your life's too hard, just go have a talk with Claude



Ken Kehoe

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15 min read · Jun 9, 2024

 326

 2



**Taking advantage of “hidden uncertainty” to
safeguard data science agents**

“Sanity Checks for Agentic Data Science”

We propose **input perturbations by taking advantage of “hidden uncertainty”** as a way to assess if agents have found stable and reliable signal or are improperly responding to noise.

Co-authors



Zach Rewolinski*



Austin Zane*



Hao Huang



Chandan Singh



Chenglong Wang



Jianfeng Gao

<https://arxiv.org/abs/2604.11003> (submitted)

* denotes equal contribution

Motivation: Coding Agents + Data Science

Agentic Data Science (ADS) Capabilities

ADS systems (e.g. Claude Code) can:

- Input: Dataset + Question
- End-to-end statistical analysis
- Generate polished reports

room

The Problem

Results can be wrong in
hard-to-detect ways.

Central Research Question:

How can we detect when an ADS conclusion is
unstable → untrustworthy?

Agent Setup & Usage

Agent Configuration

GPT-5.2-Codex with “*high*” reasoning effort.

Output Format:

Continuous 0–100 Likert scale

0 = Strong No | 100 = Strong Yes

Accessible Files

- AGENTS.md (instructions)
- Metadata (JSON format)
 - Includes research question and dataset description.
- Tabular dataset (CSV)
- Python package list

Example Research Question

“Does having children decrease the engagement in extramarital affairs?”

Our Approach: Perturbations + Stability Measures in Response Distributions

To assess the stability of the conclusions made by a data science agent, we observe how its responses vary to particular changes to its inputs.

Negative Control (NC) Perturbations

Puts the agent in a controlled environment where no relationship should be found. That is, Independently shuffling the values of each feature to destroy existing correlations.

Positive Control (PC) (PCS) Perturbations

Leave the primary signal in the data unchanged while testing sensitivity to reasonable variations.

- Expected behavior: Experienced human data scientists remain unaffected.

Goal: Characterize the empirical response distribution to detect instability.

Perturbation Types

Perturbation	Type	Targeted Failure Mode
Shuffle Feature Values	Neg. Control	Incorrect analysis conclusions
Add Non-Signal Features	PCS	Distraction by irrelevant covariates
Anonymize Feature Names	PCS	Reliance on semantic variable names
Shuffle Feature Names	PCS	Robustness to misleading names
Positive Leading Statement	PCS	Sycophancy toward user beliefs
Negative Leading Statement	PCS	Sycophancy toward user beliefs

Experimental Setup

We evaluate on 11 real datasets from the BLADE benchmark [Gu et al., 2025], a suite proposed to evaluate agentic data science abilities.

- Each dataset comes from an open-source research paper seeking to answer an associated yes/no question.

Experimental Setup:

- *Both NC & PC (PCS) distributions:* apply 5 PCS perturbations for 20 replicates each = 100 observations per dataset
- NC distribution: additional independent shuffling of feature values

Sanity Check 1: Does the Agent Find Signal?

The agent is asked a binary question, such as *"Is a lower student-teacher ratio associated with higher academic performance?"*

Sanity Check

Is the agent consistently affirmative on the real data?

- A single high response score could be a lucky run. Aggregating over many runs guards against this.
- Test whether the “**positive control**” **distribution** mean is in the "Yes" range. Since we use a Likert scale from 0–100, we test:

$$H_0 : \mu_{\text{alt}} = 50 \text{ vs. } H_1 : \mu_{\text{alt}} > 50$$

- One-sided bootstrap test w/ Phipson & Smyth correction.

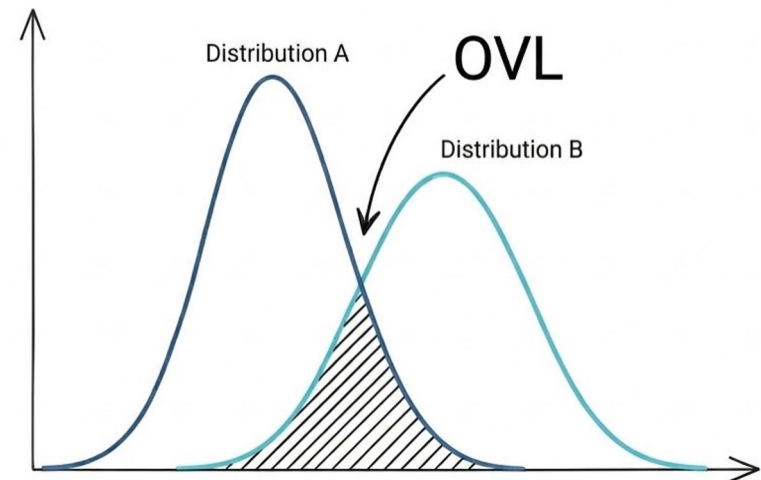
Sanity Check 2: Compare to Behavior on Noise

In data science, there often is not a clear "correct answer". Thus, we need a way to observe agent behavior in a regime with unknown ground-truth.

Sanity Check

Can the agent distinguish real signal from noise?

- Compute the **Overlap Coefficient (OVL)** between the NC and PC response densities (f_{null} , f_{alt}):
- **Low OVL** → NC & PC are meaningfully different.
- **High OVL** → NC & PC lead to similar conclusions.



Interpreting the Two Checks Together

Both checks are needed—each catches a distinct failure mode.

Overlap Amount (Sanity Check #2)

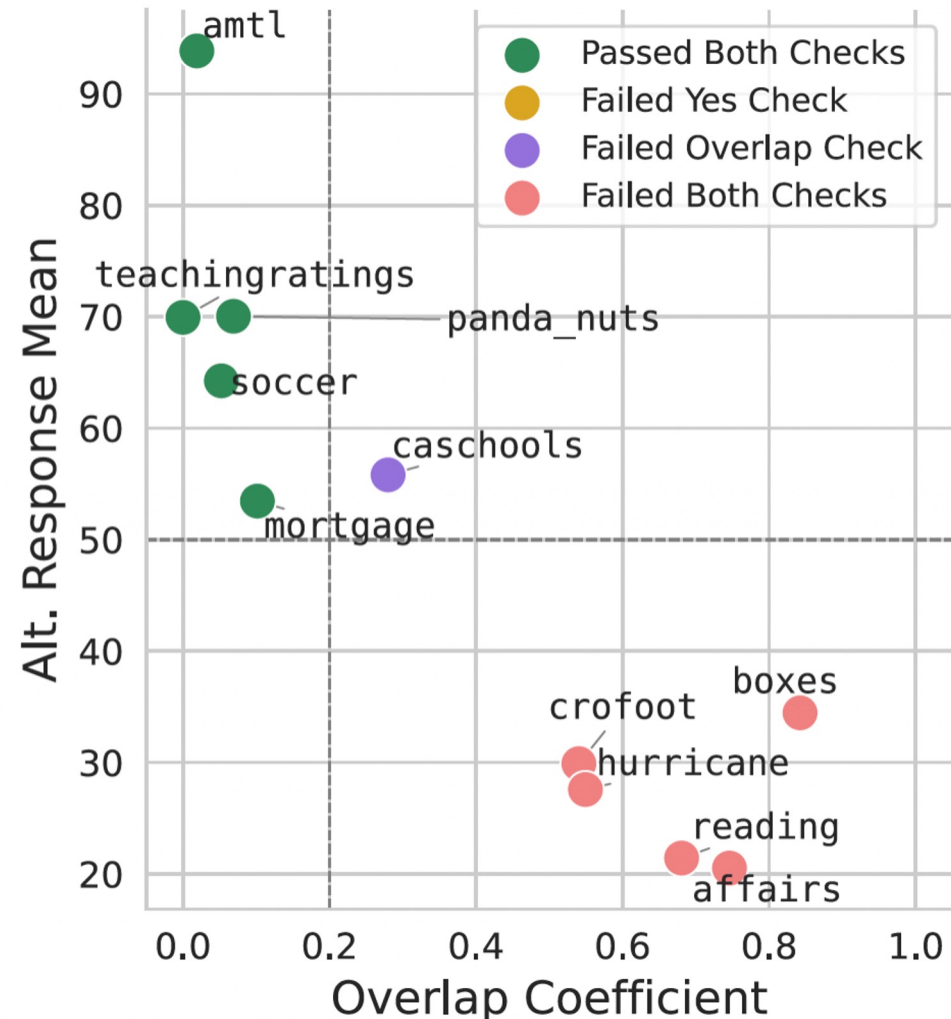
Agent's Answer (Sanity Check #1)	Overlap Amount	
	Low Overlap	High Overlap
Yes	<i>Confident yes.</i> Positive conclusion is stable.	<i>Analysis failure.</i> Positive conclusion is not grounded in the data.
No	<i>Low signal.</i> Some signal present, but not enough for a “Yes”.	<i>No signal.</i> No evidence supporting a positive conclusion.

Takeaways:

- **caschools** is the only benchmark dataset out of 11 to fail the checks
- Passing checks does not guarantee the analysis is “good”. So we have NOT proved that the agent did well for 10 datasets

“Data leakage” is possible or genAI might have been trained on the papers for which BLADE took the datasets.

More experiments on private data are needed to see how much agents rely on existing public data & analysis code.



Theoretical Foundations of PCS: much work is needed

What is called PCS-related theory?

Explicit considerations of stability or sensitivity and/or computation budget in an algorithm or procedure or under multiple generative models. “P” is covered in the generative model which is chosen for capturing reality and analytical tractability.

Existing results showing the critical role of “S” in Stat/ML theory

Bernoulli **19**(4), 2013, 1484–1500
DOI: [10.3150/13-BEJSP14](https://doi.org/10.3150/13-BEJSP14)

Stability

BIN YU

- Central Limit Theorem
- Uniform stability in ML (that bounds generalization error)
- Random matrix results
- Sensitivity bounds in causal inference

Stability is more central than Gaussian distribution to stat/ML.

Recent theoretical PCS-related works

- Behr et al (2022) shows that a theoretical version of iRF is model selection consistent under additive Boolean regression models.
- Cai and Ma (2022) on understanding effect of initialization in t-SNE cluster locations, and consistency results for clustered data
- Trillos et al (2025) studies a general “insensitivity” quantity of an estimator, and establishes a theory of sensitivity based on Weissenstein geometry, similar in spirit to the Fisher-Rao geometry.
- Luo et al (2026+) provides theory and data results to understand PCS-UQ.

PCS/VDS Resources

Software to address “C” in PCS



Veridical Flow: (v-flow) PCS-style data analysis made easy!
(Duncan et al, 2022, JOSS)



A. Agarwal



J. Duncan



R. Kapoor



C. Singh



simChef: PCS-style simulations made easy!
(Duncan et al, 2024, JOSS)

Merits: simulation guidelines (Elliott et al 2025, JOSS (to appear))



J. Duncan



C. F. Elliott



T. Tang



M. Behr



K. Kumbier

More at my website: <https://binyu.stat.berkeley.edu/> – click on code on top

MERITS of a high-quality simulation study

(Elliott et al, 2025): PCS-inspired simulation guidelines to address “C”

(Computability)

Modular: Written in self-contained and logically partitioned segments of code.

(Computability)

Efficient: Streamlined computationally and conceptually.

(Predictability)

Realistic: Faithful to the physical world.

Intuitive: Sensible to the intended audience and, in a general sense, to a reasonably comprehensive readership.

(Stability)

Transparent: Documented thoroughly and candidly.

Stable: Reproducible/replicable, and externally valid.



C.F. Elliott



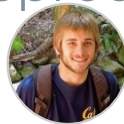
T. Tang



J. Duncan



M. Behr



K. Kumbier

PCS documentation



T. Tang

A. Kenney

Template at my website: <https://yu-group.github.io/vdocs/PCSDoc-Template.html>

1 Domain problem formulation

2 Data

3 Prediction Modeling

4 Main Results

5 Post hoc analysis

6 Conclusions

1 Domain problem formulation

What is the real-world question? This could be hypothesis-driven or discovery-based. ⓘ

This should be very high level, providing the big picture behind the study. Often this takes the form of a pre-existing hypothesis (e.g., individuals with a specific genetic mutation are more likely to have a given characteristic) or more open-ended discovery (e.g., identify mutations that are related to a given characteristic).

Insert narrative here.

↩ ↪ T π +

Why is this question interesting and important? What are the implications of better understanding this data? ⓘ

PCS workflow paper makes practicing VDS easy

To appear in a special issue on *Workflow for Applied Data Analysis*

edited by A. Gelman

PHILOSOPHICAL
TRANSACTIONS A

royalsocietypublishing.org/journal/rsta

Research



PCS Workflow for Veridical Data Science in the Age of AI

Zachary T. Rewolinski¹ and Bin Yu^{1,2,3}

¹Department of Statistics, UC Berkeley

²Department of EECS, UC Berkeley

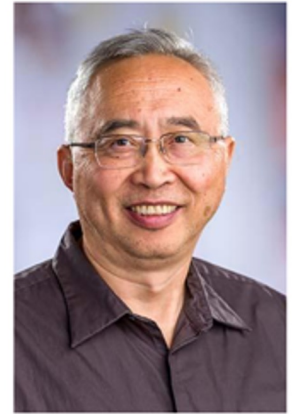
³Center for Computational Biology, UC Berkeley



<https://arxiv.org/abs/2508.00835>

PCS recommended at NCI (March, 2026)

The Steering Committee meeting of **EDRN (Early Detection Research Network)** was held in March 2026. EDRN is a research network funded by **NCI (National Cancer Institute)** and this March meeting gathered ~100 guarantee researchers and NCI project directors. EDRN Coordinating Center Principal Investigator, **Dr. Ziding Feng of Hutchinson Cancer Center, recommended that EDRN data analysis projects follow the PCS framework for Veridical Data Science.** This recommendation was well received by the attendees.



Ziding Feng

PCS is essential to BW and AI Start-up Traversal



*"In **AIA Labs at Bridgewater**, we use artificial intelligence and machine learning to understand the global economy and manage investments in a noisy, adversarial, high-stakes environment. Standard statistical inference techniques make strong assumptions for the sake of analysis; when applied blindly, without verifying these assumptions or accounting for researcher judgement calls, they are dangerous and convey a false sense of confidence. **The PCS framework influences how we develop, apply, and govern our methods by forcing us to interrogate both collected data and the theoretical data-generating process as we develop algorithms.**"*

**– Jas Sekhon, Chief Scientist and AI Head (until 3/26)
(Now Chief Strategy Officer, Google DeepMind)**

"Traversal builds an AI Site Reliability Engineer (AI-SRE) that helps companies like **American Express, Pepsi, and DigitalOcean** diagnose outages by searching petabytes of telemetry and code to deliver fast, trustworthy root causes and resolutions.

In these high-pressure settings—where signals are very weak and the search space is enormous—the PCS framework is essential.

First get accuracy (P). Then speed (C). Then stability (S). That's how we develop at Traversal AI.

– Raaz Dwivedi, co-founder, CTO; Assist. Prof. Cornell Tech

<https://fortune.com/2025/06/18/traversal-emerges-from-stealth-with-48-million-from-sequoia-and-kleiner-perkins-to-reimagine-site-reliability-in-the-ai-era/>

Fortune 6/25

Traversal emerges from stealth with \$48 million from Sequoia and Kleiner Perkins to reimagine site reliability in the AI era

By Allie Garfinkle

Senior Finance Reporter And Author Of Term Sheet



PCS recommendations for any data project

Concretely and “**in-context**”,

- For problem formulation, try more than one
- For data cleaning, at least keep two copies of cleaned data
- For EDA and reporting, try different graphics parameters, and data perturbations
- For modeling step, it values multiple good or pred-checked models, moving away from “true” model framing; and it provides principled ways to aggregate good models (incl. those from data perturbations)

Last but not least: ways to use the VDS book in your class

- Send the free version vdsbook.com link to your class with some comments
- Keep your curriculum, but do a class prediction project Kaggle style following PCS
- Use part of the book: data projects, chapter on problem formulation, chapter on data preparation, all first 4 chapters that are not in typical stat books
- Use the whole book

In my PhD and MA classes (stat 215A and stat 214 at UCB) on Applied stats+ML follow PCS. Students read the book chapters and submit book summaries, and my own lectures use VDS book and other materials from research and other sources.

More ideas and experiences on teaching VDS can be found in

Rosenberg, J., Bonvini, M., Bray, A., Gong, R., and Yu, B. (2026).



"Teaching Veridical Data Science: A Panel Interview with Matteo Bonvini, Andrew Bray, Ruobin Gong and Bin Yu," (2026)

Journal of Statistics and Data Science Education, Vol 34(2).

<https://www.tandfonline.com/doi/full/10.1080/26939169.2026.2632565>

This project was started with a contributed session at 2025 JSM on VDS education organized by Gong and Bonvini.

PCS is an evolving frontier of statistics

- PCS principles work for a DSLC or AI workflow and embrace pluralism
- Documentation is an integral part
- Many open areas: multi-modality, dynamic data, PCS theory, ...

More VDS workshops: check Bin's website

Past 4 workshops:

[Paris Workshop on Veridical AI](#)

May 28-29, 2026, Centre Borelli

[Veridical Data Science in Biology](#)

July 11, 2025, QB3, UC Berkeley

[Rome Workshop on Veridical Data Science](#)

June 20, 2025, Sapienza University, Rome

[**Inaugural Berkeley-Stanford Workshop on Veridical Data Science at UC Berkeley**](#) (May 31, 2024) (talk videos available)

**Thanks to Le Cam and
Speed, other mentors,
and all my collaborators!**

Thank you!

Questions?



Published papers

B. Yu (2013). [Stability](#). *Bernoulli*.

S. Basu, K.Kumbier, B. Brown, B. Yu (2018), [Iterative random forests](#). *PNAS*.

B. Yu and K. Kumbier (2020). [Veridical data science](#). *PNAS*.

B. Yu (2023) [What is uncertainty in today's practice of data science?](#) *J.Econometrics*.

B. Yu and R. Barter (2024). Veridical data science: the practice of responsible data analysis and decision making. *MIT Press* (online free version at vdsbook.com).

T. Tang, Y. Zhang, A. Kenney, ..., B. Yu, Arul Chinnaiyan (2025). [A simplified MyProstateScore2 for high-grade prostate cancer](#). *Cancer Biomarkers*.

Q. Wang*, T. M. Tang*, ..., B. Yu*, E. Ashley* (2025). [Epistasis regulates genetic control of cardiac hypertrophy](#). *Nature Cardiovascular Research* ([Code](#)) ([PCS documentation](#))

Recent papers

Z. Rewolinski and B. Yu (2025) PCS workflow for veridical data science in the age of AI . <https://arxiv.org/abs/2508.00835> (to appear in *Philosophical Trans. A of Royal Society*)

I. Azizi, J. Bodik, J. Heiss, B. Yu (2026). CLEAR: Calibrated Learning for Epistemic and Aleatoric Risk. ICLR.

A. Alaa and B. Yu (2024) Veridical Data Science for Medical Foundation Models. <https://arxiv.org/abs/2409.10580>

A. Agarwal, M. Xiao, R. Barter, B. Fu, O. Ronnen and B. Yu (2025). PCS-UQ: Uncertainty Quantification via the Predictability-Computability-Stability Framework <https://arxiv.org/abs/2505.08784> (revised for PNAS)

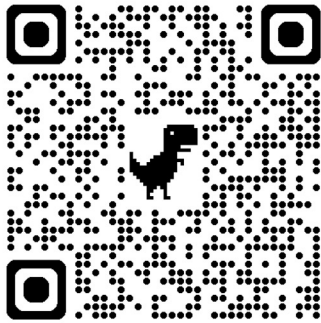
R. Ma*, X. Li, J. Hu, B. Yu* (2025). Uncovering smooth structures in single-cell data with PCS-guided neighbor embeddings. <https://arxiv.org/abs/2506.22228> (accepted by *Patterns*)

Check out simons.berkeley.edu for future, current, and past workshop talk videos on AI, ML, and Theory of Computing (TCS)

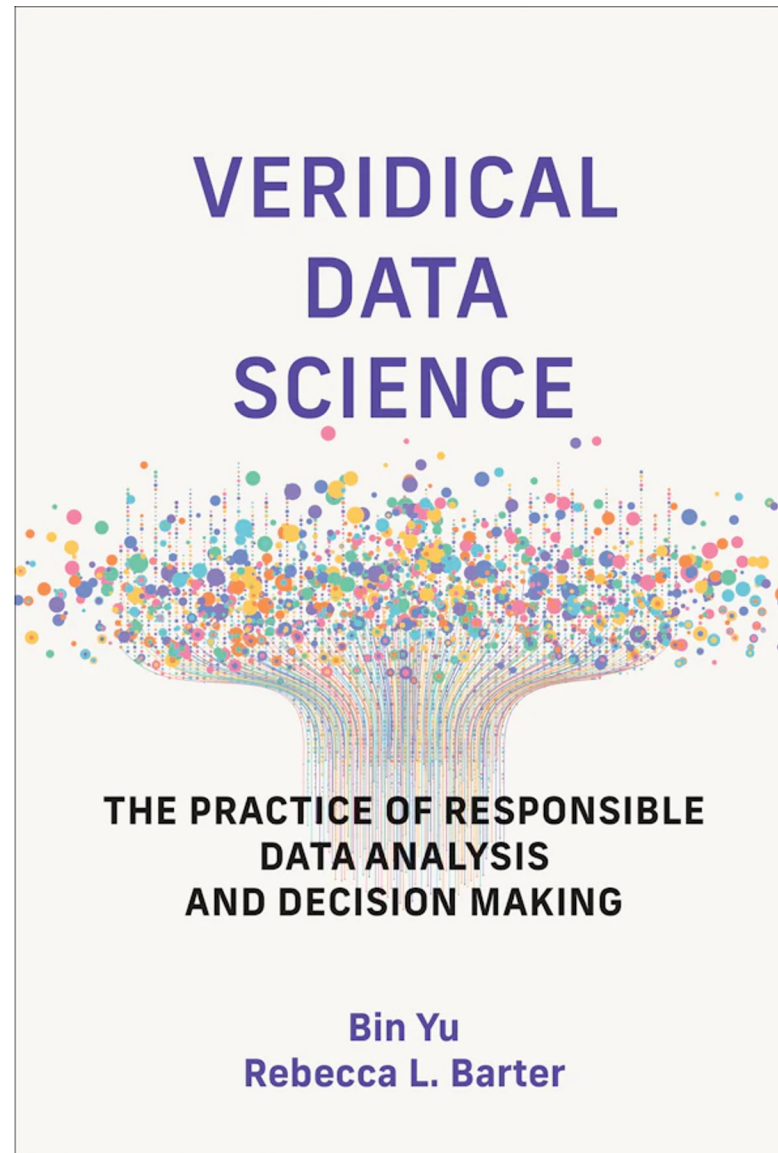
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